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Performance measurement of reading teaching-learning Business Processes: case of whole-word and syllabic reading methods in primary schools

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Abstract

In the educational field, Key Performance Indicators (KPIs) aim to provide an opportunity to evaluate the performance of employees, teachers' skills and pedagogy, students' results, learning objects, educational institutions and teaching-learning methods. In this research work, we focus on studying the performance of reading teaching-learning methods, specifically whole-word and syllabic languages learning methods. More precisely, we propose a Business Process Management (BPM) based approach for performance measurement of the corresponding business processes of the both methods. We elaborated the reading teaching-learning business processes using Business Process Model and Notation (BPMN), which is a widely adopted standard for formalizing business process modeling. We prepared a set of KPI for the different activities and for the whole process. We used these KPIs to gauge the performance of each business process and make comparisons between them. To validate our approach, we launched a dedicated questionnaire to primary school teachers in the third grade. By using this questionnaire, we were able to evaluate the overall performance of each reading process. Our findings indicate that the whole-word method is the most effective teaching method compared to the other.

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Keywords: KPI, BPM, education, performance, teaching-learning method, business process, whole-word, syllabic reading method.

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1. Introduction

Various techniques have been used worldwide to study teaching-learning processes. Numerous researchers have indeed conducted studies on the management of teaching-learning Business Processes (BPs). In this context, Business Process Management (BPM) is a management discipline that focused on the use of BPs as an important contributor to the achievement of an organization's goals through improvement, continuous performance management and governance of its essential BPs [1]. A plethora of BPs could be considered in the field of education, such as the process of managing the planning and coordination of teachers' schedules, implementing online learning management systems, developing and implementing continuous training programs for teachers, and planning and implementing school support services and learning assistance for students. Currently, multiple methods have been suggested for assessing the performance of BPs, one of which is the management of Key Performance Indicators (KPIs). This method is crucial for enhancing BPs, as well as aiding organizations in defining and measuring the progress made towards attaining their objectives [2]. In the realm of education, multiple studies have concentrated on establishing and defining various KPIs. KPIs allow measuring the progress and performance of students, teachers, and educational institutions. Several studies have identified several important KPIs, such as exam success rates, graduation rates, student and teacher satisfaction, student attendance and punctuality rates, as well as student success rates in pursuing higher education or in entering the workforce. Educational institutions make use of these KPIs to assess their own performance and identify areas that need improvement. They can also be used to compare the performance of educational institutions against each other. For instance, authors, in [3], proposed a study based on models of excellent teaching. The theoretical framework of this study presents certain requirements that are generally accepted and recommended for excellent teaching or key performance indicators to become an excellent teacher. Primary schools place a strong emphasis on language mastery, specifically reading proficiency. Reading is a complex skill that requires the coordination of decoding – a resource for word recognition through retrieving the sounds represented by letters (phonemes) – and processes shared with oral language comprehension [4]. Consequently, learning to read is a significant objective, which necessitates a thoughtful selection of teaching-learning methods for this task. "The common methods of teaching reading in primary classes are the phonic method, the syllabic method, the whole-word and whole-sentence (look and say) method, and the eclectic method [5]. We have selected two of the most used methods in teaching: the whole-word reading method and the syllabic reading method. The official program in Tunisia incorporates two reading methods, namely, the syllabic method and the whole-word method. Educators are facing difficulty in determining which of the two methods is the best. The overall aim of this paper is to investigate which reading teaching and learning method is more effective. Our main challenge is therefore to seek a solution that can successfully incorporate KPIs into BPs, thereby enabling the evaluation of their performance. Additionally, we conduct a comparative study between business processes based on the obtained KPIs. From this perspective, we provide a description and comparison of the BPs (syllabic reading process and whole-word reading process) based on KPIs, with the aim of determining which of these two processes is the most performant and effective. In our paper, we have structured our content by following a specific plan. We first explore business process management in the context of education, and then focus on KPIs in the same context. Our third section is dedicated to an approach for measuring the performance of language learning BPs. To illustrate our approach, we present a case study of a primary school's third year. In the fifth section, we review related work on the topic. Finally, we conclude our article by summarizing our main findings and suggesting future research directions.

2. Business process management in the context of education

Nowadays, a BP is a very important concept in the field of information systems. BPM brings gradual improvements to BPs using various methodologies, techniques, and tools to ensure the organization's competitive position in the market. In this section, we define BP and BPM, and outline the importance of BPM in the context of education.

2.1 BPM and BP definition

Several definitions of the term BPM are presented in the literature. Indeed, BPM is defined as "the art and science

of overseeing how work is performed in an organization to ensure consistent results and to take advantage of improvement opportunities.” [6]. BPM provides an overview of organizational resources and capabilities and assigns them to tasks and activities within the company’s value creation processes [7]. There are many other definitions, but in essence, they are all the same. A BP is a partially ordered set of business activities that can be executed to achieve a specific objective of a company or part of a company in order to reach a desired result [8]. Therefore, a BP is a set of coordinated activities that are designed to produce results for internal or external customers. It is executed by human actors or automated systems, using different resources.

2.2 Educational bpm

In today’s fast-paced world, BPM is increasingly being used to support and manage complex processes within organizations. More precisely, in educational systems, it can be used to promote teaching-learning processes with the advantage of aligning actual tasks with training tasks [9]. The goal is to associate business activities with learning tasks and to use available BPM techniques for supporting, advising, simulating, and monitoring learning activities. As a result, the potential of BPM techniques has been leveraged to support effective and realistic learning and teaching activities. BPs play an important role in improving the efficiency and effectiveness of educational activities. They help educational institutions to better manage their resources and improve student experiences.

3 KPI in the context of education

In recent years, research on KPI management has experienced exponential growth, resulting in a multitude of heterogeneous approaches addressing various aspects. In this section, we establish the different definitions of KPI, KPIs in the context of BPM, and finally KPIs in the context of education.

3.1 KPI definition

The diversity of proposed definitions of KPI is evident in a few basic properties that are present in many approaches, such as the KPI identifier, its name, and its text description provided in natural language. Indeed, KPIs are the indicators that focus on aspects of organizational performance that are most critical to the current and future success of the organization [10]. Besides, it is a measure that evaluates how well a company executes its strategic vision. The term strategic vision refers to how an interactive strategy is integrated into an overall business strategy.

3.2 KPI in BPM

KPIs have become the standard approach to monitoring and managing a company’s performance [11]. As such, BPM has brought a revolutionary alternative to the traditional way of managing a business. Instead of establishing KPIs for specific areas of the organization, they should be derived from its business processes. Furthermore, in the BPM field, performance management is a very important link. The KPI method has been implemented in performance evaluation by various types of companies. This should be attributed to two striking characteristics of KPIs [12]. First, KPIs emphasize the importance of performance indicators that align with the organization’s development strategy. Second, KPIs focus on identifying and addressing problems within a specific period, promoting accountability and helping to ensure that the organization stays on track towards its goals. In BPM, the ability to measure KPIs is crucial. Without this capability, BPs cannot be optimized intelligently or in response to strategic events to create valid new BPs. Therefore, the incorporation of the KPI into the BP model has enhanced the performance of the BP and addressed the pressing requirement of the company for effective business process management.

3.3 KPI in the educational context

According to [13], KPIs are used to evaluate organizational performance against strategic and operational objectives that allow for quantifiable measures reflecting essential success factors of a university. In the higher education sector, strategic and operational performance indicators are determined by top management in consultation with financial organizations such as Deloitte and Touche, Ernst and Young, and Price Waterhouse Coopers. KPIs play a crucial role in improving performance within education and have gained increased importance after the introduction of performance-based funding in higher education. Approach for performance measurement of Languages learning Business Processes

Our approach consists of integrating KPIs at the BPM level to improve the teaching-learning processes, specifically languages reading BPs. In this research paper, we consider reading teaching-learning method's. A teaching-learning method is a procedure, a classroom strategy, an ordered sequence of techniques. It is about how the teaching is to be conducted. It can be described in terms such as “first you do this, and then do this . . .”. The selection of the methods depends on an analysis of the most used ones in primary schools to teach foreign languages (French). These methods will be next modeled using selected Business Process Modeling Languages (1st Step) and we will identify reading teaching-learning KPIs for each of these methods (2nd Step). The BPs will be executed and KPI will be measured (3rd Step), using questionnaire in order to analyze the performance of learners. Finally, we will construct formulas for the measurement of BPs' performance (4th Step). Our approach is thus composed of four main steps as described in Figure 1.

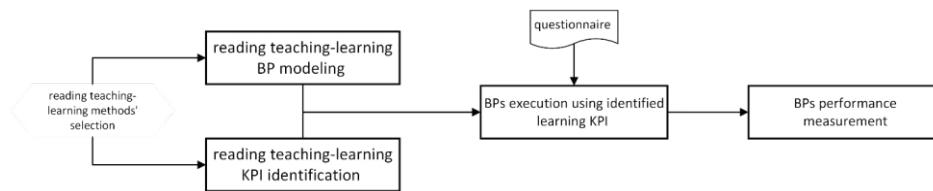


Fig. 1. Languages learning BP performance measurement approach using KPIs.

3.4 Step 1: modeling of languages' learning business processes

In this step, our aim is to model the languages' learning BPs. We have selected the most used methods . . . Methods of teaching reading tend to be primary concerned with the roles of the teacher and students. To simplify these processes, we have considered that the main role is the teacher. Both BPs are modeled in BPMN. We used the BPMN modeling language because it allows visually representing business processes in a clear and understandable way for the different stakeholders involved in their design and execution.

3.4.1 Syllabic reading method BP

The syllables are taught first, followed with words, and sentences and then reading. The syllabic reading BP unfolds as follows. First of all, learners should be identified by teachers. Next, teachers present the letters and their corresponding sounds to learners and mention if learners could identify and pronounce correctly or incorrectly the different letters. Then, teachers specify if the learner is able or not to assemble the letters and to form and pronounce syllables, and finally to be able to read and to understand correctly words.

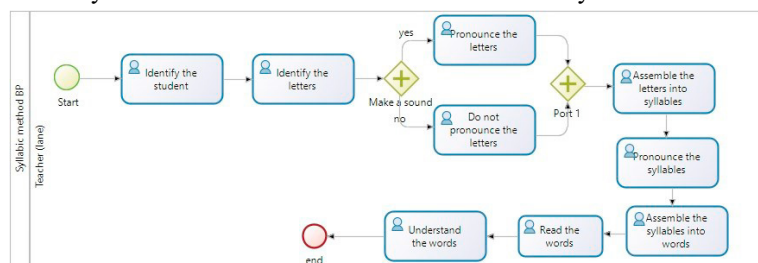


Fig. 2. Syllabic learning method BP.

3.4.2 Whole-word method BP

First of all, learners should be identified by teachers. The ‘Whole-Word’ method is also called as ‘look-and-Say’ or ‘see-and-say’ method or global method. The ‘whole-world’ method begins with a word. The words may be the content and / or structural words. Instead of teaching individual sounds, this BP approach begins by directly presenting fully memorized words, implicitly conveying the sounds that constitute them. The learning of these words is followed with syllables and letters, after which the actual reading begins. The learners are expected to look at the general appearance of the words and are expected to memorize the sounds that should be spoken.

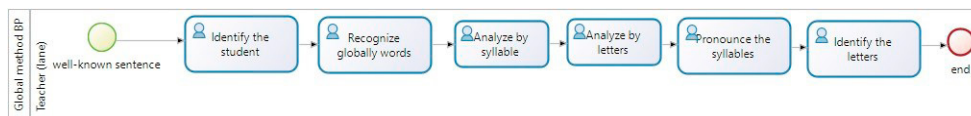


Fig. 3. Whole-word learning method BP.

These two processes were validated by an expert in the educational field, who was provided with an explanation of the execution of both processes.

3.5 Step 2: languages learning KPI identification – extraction

Learning the correct pronunciation of words can be challenging and the level of difficulty can vary. It is common to make mistakes and not always pronounce words correctly while attempting to learn the correct way of saying them. In fact, some sounds seem to be difficult to pronounce than others, or there are words that contain unpronounced syllables or letters. Based on this point, we describe the different reading KPIs. Authors, in [14], have proposed a set of multiple types of Distances from the Expected Performance (DEP) that measure the expected student's performances in the expected level.

- DEP in reading by direct route (addressing), that is, the recognition of words without relying on the processing of their grapho-phonetic components, but through the direct matching of the written word and its visual representation in memory.
- DEP in reading by indirect route (assembly), that is, through the application of correspondence rules between graphemes and phonemes.
- DEP in reading by an undetermined route: the student must read words in context, in sentences equivalent to those studied during the reading session. They may use the indirect and/or direct route, with the help of context.
- DEP in comprehension: we orally ask the student questions about the understanding of the text studied.

Authors, in [15] highlighted the various ways in which we can analyze and provide information on the activity of learners in a course. This includes being able to visualize the level of learner participation, check the overall progress percentage of each individual learner, and review the activities and resources completed by each learner. Overall, analyzing activity and providing information to learners and teachers can be a powerful way to support learning and ensure that each learner reaches their full potential. Based on the research works of [14] and [15], we have identified the following KPIs:

- KPI 1: The learner reads accurately with good pronunciation of sounds.
- KPI 2: The learner demonstrates good awareness of the phonemes of each word.
- KPI 3: The student has isolated reading of syllables.
- KPI 4: The learner is participating well in class.
- KPI 5: The student's progression is impressive.
- KPI 6: The student has isolated reading of words.
- KPI 7: The student has identified the letter(s) correctly.
- KPI 8: The student has understood the meaning of words.

- KPI 9: The student has assembled the letters into syllables correctly.
- KPI 10: The student has assembled the syllables into words correctly.
- KPI 11: The student has corrected any reading errors.
- KPI 12: The student has checked that the text makes sense.

3.5.1 KPI 13: The student has listened to and discussed poems and stories in order to read them independently. *Languages learning KPI for activities*

We have assigned each KPI to a BP activity as presented in Table 1.

Table 1. Assignment of KPIs to process activities.

Activity	KPI
Identify letters	KPI 7
Pronounce letters	KPI 1 + KPI 11
Assemble letters into syllables	KPI 9
Pronounce syllables	KPI 3 + KPI 11
Assemble syllables into words	KPI 10 + KPI 2
Read words	KPI 6 + KPI 11
Understand words	KPI 8
recognize globally words	KPI 2 + KPI 8 + KPI 10
Analyze by syllables	KPI 9
Analyze by letters	KPI 7

3.5.2 Languages learning KPI for the whole BP

Regarding the learning ... All process activities include KPI 4, KPI 5, KPI 12 and KPI 13.

3.6 Step 3: business process execution using identified learning KPI

Questionnaire preparation

We have proposed a questionnaire that contains a series of 17 questions representing the selected KPIs. The questions in this questionnaire make use of the Likert quality scale, which allows responders to select from a range of possible answers, including Very Poor, Poor, Fair, Good, and Excellent. The target group for this questionnaire are teachers, and it is presented in Google Form. The questionnaire is accessible through the following link: shorturl.at/cyT12. It is important to note that the language used in the questionnaire is French.

Framework for BPs execution

Several BPMS (Business Process Management Systems) exist on the market to support the business process management lifecycle. Bonita¹ BPMS is considered as one of the most widely used systems in business process management and is distributed under an open-source license. Bonita allows deployment via a standalone web administration platform designed to integrate with existing environments in organizations. It is a database that the specialized community is still working on and improving [16]. For these reasons, we have chosen Bonita as our business process management tool.

Attributes set preparation

We created a database entitled "Texte," in which we added eighteen attributes, including five attributes "phrase", "letter", "syllable", "word" and "sounds", that contain information about the sentence to be learned by the learner, and thirteen attributes, which are from "KPI 1" to "KPI 13", that indicate the teacher's evaluation of the students' level. The data is a record of the sentences to be learned and the corresponding evaluation provided by the teacher for each student, which is a direct outcome of the execution of our process. Therefore, it reflects a historical perspective of the learning process.

Forms

After implementing the process and creating specific forms for our application example using the tools provided by the Bonita Business Process Management system, we executed the BPs. Nine forms were created for the syllabic reading method, and six forms for the whole-word reading method. One form for each reading method is presented in Figure 4. The learner must break down the sentence and identify all the letters in that sentence. The teacher can assess the learner's level of comprehension through a set of questions presented in Figure 4(a). The learner must pronounce the syllables where the teacher can assess their level of syllable pronunciation (Figure 4.(b)).

Fig. 4. (a) The task form for the syllabic method "Identify the letters"; (b) The task form for the whole-word method "Identify the letters".

3.7 Step 4: Business processes performance measurement

To measure the performance of each reading process (whole-word and syllabic), we establish the following mathematical formulas:

$$\text{Global performance} = \frac{\sum \text{Performance of each instance}}{\text{Number of instances}} \quad (1)$$

with:

$$\text{Performance of each instance} = \frac{\text{Activity Performance}}{\sum \text{Activity KPI}} + \text{Business process performance} \quad (2)$$

$$\text{Activity Performance} = \sum_{\text{For each activity}} \text{Number of KPIs per activity} \quad (3)$$

$$\text{Business process performance} = \sum_{\text{At the process level}} \text{KPI} \quad (4)$$

To obtain the Global performance, we add up the individual performances of each instance and divide this sum by the number of instances. Then, the performance of each instance is calculated by dividing the activity performance by the sum of activities and adding the corresponding business process performance. Next, the activity performance represents the performance measurement of a specific activity. It is obtained by adding the values of key performance indicators (KPIs) for that activity and dividing this sum by the total number of KPIs for that activity. The process-level performance is derived by summing up the values of key performance indicators (KPIs) at the process level.

4 Case study: Primary School – 3rd year

Our survey was conducted at the primary school 'Brahim Ezzaher' located in Sbiba in Tunisia. We distributed the questionnaire to two French teachers in the third year of primary school. We present a comparative study between the two reading processes to evaluate the overall performance of each method. In a sample of 10 students from each reading process, we noticed that using the whole language approach, six students received a very good evaluation from the teachers. Additionally, using the phonics-based approach, we found that only two students received a very good evaluation. The most successful reading process is the one with the highest overall performance. Based on the results obtained, we noticed that the whole language approach (= 34.66666667) is more effective than the phonics- based approach (= 13.01111111). Therefore, we found that the whole language reading method allows learners to reliably master discourse and reflection before mastering the necessary pronunciation skills within the context of understanding the French language. Additionally, whole language reading develops visual attention and memory skills.

5 Discussion

Since our work describes KPIs for business processes in the context of education, several research studies exist in this field. In this section, we will examine some of these studies: Ilango Sivaraman et al. presented a method for developing KPIs for the mission and vision statements of an engineering college in Muscat, Oman. In this context, KPIs provide parameters for measuring the quality and standards of the institute. Their goal is to develop secondary school students into responsible and competent engineers. They considered that the performance measurement elements are incorporated in the mission and vision statements [17]. Vukšić et al. proposed a BPM approach based on the use of simulation models to enable organizations to maintain and improve their operational performance [18]. Wiechetek et al. presented a BPM deployment model for logistics students at Maria Curie-Skłodowska University (MCSU) in Poland. They established that an ideal educational software should cover a wide range of popular models and standards and should reflect the entire BPM lifecycle [19]. The presented works could compare and evaluate the existing KPIs and BPM approaches used in education against some new or improved ones, propose a new methodology or tool for developing and measuring KPIs for business processes in education and in this work, we were interested in executing and comparing reading methods.

6 Conclusion

This paper focuses on the topic of teaching reading performance. Our goal is to compare the two selected business processes for their efficiency, effectiveness, and overall performance in achieving the desired outcomes (syllabic reading process and whole-word reading process) based on the KPIs obtained to have the most efficient process to ensure a certain reliability in teaching. To achieve this objective, we first focused on presenting a model of student evaluation processes by the teacher which in turn integrates KPIs in BPM, those that best meet our objectives and facilitate the definition of our approach. In the second step, we presented our approach in which we identified the reading KPIs for each activity and student evaluation process by the teacher to measure their performance. This research field is vast. As a continuation of this work, we propose the following research perspectives: Studying other reading methods such as mixed, ideovisual, etc., and addressing other types of more complex processes.

References

- [1] Jeston, J. (2014) "Business process management: practical guidelines to successful implementations." Routledge.
- [2] Arora, A. and Kaur, S. (2015) "Performance assessment model for management educators based on kpi." In *International conference on technology and business management* volume 23.
- [3] Cabrera, A. F., Colbeck, C. L., and Terenzini, P. T. (2001) "Developing performance indicators for assessing classroom teaching practices and student learning." *Research in higher education* 42 (3):327–352.

- [4] Giménez, A., Sánchez, A., Flores, A., and Luque, J. L. (2023) “Teachers’ opinions about the teaching of reading in Spain.” *Educational Psychology* **29** (1):65–73.
- [5] Batirovna, M. D. (2022) “The methodic of development the innovation technologist.” *Galaxy International Interdisciplinary Research Journal* **10** (2):266–268.
- [6] Bandara, W. and Opsahl, H. (2017) “Developing organization-wide bpm capabilities in an sme: the approaches used, challenges and outcomes.” *Journal of Information Technology Teaching Cases* **7** (2):92–113.
- [7] Fischer, M., Imgrund, F., Janiesch, C., and Winkermann, A. (2020) “Strategy archetypes for digital transformation: Defining meta objectives using business process management.” *Information and Management* **57** (5):103262.
- [8] Aguilar-Saven, R. S. (2004) “Business process modelling: Review and framework.” *International Journal of production economics* **90** (2):129–149.
- [9] Bakki, A., Oubahssi, L., Laghouaouta, Y., and George, S. (2023) “Bpmn4mooc : A bpmn extension for the design of connectivist mooc.” *Interactive Learning Environments*, pages 1–23.
- [10] Parmenter, D. (2015) “Key performance indicators: developing, implementing, and using winning KPIs.” John Wiley and Sons.
- [11] Repa, V. (2019) “Deriving key performance indicators from business process model.” In *International Conference on Business Informatics Research*, pages 148–162. Springer.
- [12] Pan, W. and Wei, H. (2012) “Research on key performance indicator (kpi) of business process.” In *2012 Second International Conference on Business Computing and Global Informatization*, pages 151–154. IEEE.
- [13] Kairuz, T., Andries, L., Nickloes, T., and Truter, I. (2016) “Consequences of kpis and performance management in higher education.” *International Journal of Educational Management*.
- [14] Murillo, A. (2010) “Le choix du niveau de difficulté des tâches scolaires: des marges de manœuvre limitées pour les enseignants.” *Carrefours de l’Education* (1):77–92.
- [15] Safsouf, Y. (2021) “Contribution à l’élaboration de modèles décisionnels pour l’amélioration de la réussite des apprenants dans un système d’apprentissage en ligne.” PhD thesis, Université de Bretagne Sud.
- [16] Araújo, W. J. d. and Gomes, T. A. (2023) “Evaluation of business process management systems (bpms) : analysis of bizagi and bonita soft- wares.” *RDBCI : Revista Digital de Biblioteconomia e Ciência da Informação*, pages 20.
- [17] Ilango Sivaraman, D., Al Balushi, A., and Rao, D. (2014) “Developing key performance indicators from mission and vision statements of an engineering college in Oman.” *Global Journal of Management And Business Research*.
- [18] Vukšić, V. B., Tomicic-Pupek, K., and Bach, M. P. (2018) “Simulation based business processes analysis in higher education institutions.” In *2018 41st International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO)*, pages 1266–1271. IEEE.
- [19] Wiechetek, Ł., Medrek, M., and Banas, J. (2017) “Business process management in higher education. the case of students of logistics.” *Problemy Zarzadzania*, (4/2017 (71)):146–164.